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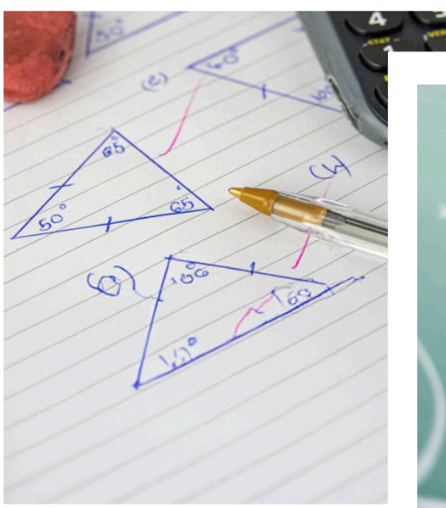
Quantitative Aptitude

Number System

STUDY NOTES

Divisibility Rule



Divisibility Rule

Rules by the help of which we can check the divisibility or factors of a number without actual division:

Divisibility by 1:

1 is not divisible by any number except 1 but 1 is a universal factor.

Divisibility by 2:

Numbers whose last digit is either even or zero are divisible by 2.

Ex; 2, 28, 78, 10 etc.

Divisibility by 3:

If the sum of digits of a number is divisible by 3, then the number is also divisible by 3.

Ex; 72, 642, 96 etc.

Example 1: A number 9876X3 is divisible by 3. What are the possible values of X?

Solution:

Given:

The number = 9876 X 3

Concept:

A number is divisible by 3 if the sum of the digits of the number is divisible by 3

Calculation:

⇒ 9876 X 3

⇒ The sum of the digits of the number = 9 + 8 + 7 + 6 + X + 3 = 33 + X

⇒ The possible value of X = 0, 3, 6, and 9

∴ The correct answer is "0, 3, 6, and 9".

Divisibility by 4:

If the last two digits of a number are divisible by 4 or the number having two or more zeros at the end is divisible by 4.

Ex; 428, 772, 700 etc. here last two digits are 28, 72, 00 which are divisible by 4.

Example 2: Which of the following numbers is completely divisible by 4?

1) **97634**
2) **47654**
3) **83544**
4) **31459**

Solution:

If the last two digits of a number are divisible by 4, then the number is divisible by 4.

1) 97634 - 34

34/4 - 17/2 Not Divisible

2) 47654 - 54

54/4 - 27/2 Not Divisible

3) 83544

44 - 44/4 - 11 divisible

4) 31459

59 - 59/4 - Not Divisible

∴ The correct answer is 83544.

Divisibility by 5:

If the last digit of a number is 5 or 0, then the number is divisible by 5.

Ex; 50, 105, 100 etc.

Example 3: Which of the following numbers is completely divisible by 5?

1) **6734**
2) **454**
3) **540**
4) **359**

Solution:

If the last digit of a number is 5 or 0, then the number is divisible by 5.

Only option 3) 540 has the unit digit 0 hence 540 is divisible by 5.

∴ The correct answer is 540.

Divisibility by 6:

If a number is divisible by 2 and 3, then the number is also divisible by 6.

Ex; 954, 732 etc.

Example 4: If the number 327AB is divisible by 6, then find the smallest value of the natural numbers A and B.

Solution:

Given:

A number 327AB is given which is divisible by 6

Concept Used:

Divisibility Rule of 6 - Numbers which are divisible by both 2 and 3 are divisible by 6.

That is, if the last digit of the given number is even and the sum of its digits is a multiple of 3, then the given number is also a multiple of 6.

Calculation:

According to the question

327AB is divisible by 6.

That means, the number must be divisible by both 2 and 3.

According to the divisibility rule of 2, the units digit of the number should be even i.e. 0, 2, 4, 6, and 8.

Here, we need the smallest value so we take B = 2.

Now, According to the divisibility rule of 3 the sum of the digits of the number should be a multiple of 3.

So, $3 + 2 + 7 + A + B = 12 + A + 2$

$\Rightarrow 14 + A$

The nearest multiple of 3 is 15.

So, $14 + A = 15$

$\Rightarrow A = 15 - 14$

$\Rightarrow A = 1$

The smallest value of the natural numbers A and B is 1 and 2.

∴ The correct answer is A = 1 and B = 2.

Divisibility by 7:

If the difference between twice the unit digit of the given number and the remaining part of the given number is a multiple of 7 or it is equal to 0, the whole number is divisible by 7.

Ex; For 203, double the last digit (3 * 2 = 6) and subtract it from the rest of the number (20 - 6 = 14), which is divisible by 7.

Example 5: Which of the following numbers is divisible by 7?

1) **3739**
2) **3661**
3) **3659**
4) **3915**

Solution:

Given:

Divisibility of 7

Concept used

The divisibility rule of 7 states that

Pick the last digit of a number, multiply it by 2,

Subtract the product from the rest of the number to its left. If the difference is 0 or a multiple of 7, then the given number is divisible by 7

Calculations:

According to the concept,

We will check if the numbers are divisible by 7

⇒ 3739 = 373 - 9 × 2 = 373 - 18 = 355 [not a multiple of 7]

⇒ 3661 = 366 - 1 × 2 = 364 [multiple of 7]

⇒ 3659 = 365 - 9 × 2 = 365 - 18 = 347 [not a multiple of 7]

⇒ 3915 = 391 - 5 × 2 = 391 - 10 = 381 [not a multiple of 7]

The number divisible by 7 is 3661

∴ **The correct answer is 3661.**

Divisibility by 8:

If the last three digits of a number are divisible by 8 or the last three digits of a number are zeros, then the number is divisible by 8. **Ex;** 8432, 5000 etc.

Example 6: Find the smallest number that can be subtracted from 148109326 so that it becomes divisible by 8.

Solution:

Given:

Divisibility rule of 8

Concept used:

Divisibility rule of 8: When the number made by the last 3 digits of a number is divisible by 8 then the number is also divisible by 8.

Apart from this if the last three or more digits of a number are zero then the number is divisible by 8.

Calculations:

Considering the given number:

The last 3 digits are 326 and so, for it to be completely divisible by 8 we have to deduct 6 from the given number.

⇒ (148109326 - 6) / 8

⇒ 148109320 / 8

⇒ 1763655

∴ **The correct answer is 6.**

Divisibility by 9:

If the sum of all the digits of a number is divisible by 9, then the number is also divisible by 9.

Ex; 8946, 765 etc.

Example 7: If the number 18345mn is divisible by 9, then find the minimum value of (m+n).

Solution:

Given:

Number = 18345 mn

Concept used:

For a number to be divisible by 9 the sum of digits must be divisible by 9

Calculation:

Sum of digits = 1 + 8 + 3 + 4 + 5 + m + n

⇒ Sum of digits = 21 + m + n

For min value of m + n

21 + m + n = 27

m + n = 6

∴ **The correct answer is (m + n) = 6.**

Digital Sum

It is a simple way to check if a number is divisible by 9 or 3. Just add up all the digits of the number. If the sum is divisible by 9, then the number is also divisible by 9. If the sum is divisible by 3, then the number is also divisible by 3.

It can be used to determine whether a number is divisible by 9 or 3 without performing the actual division.

Divisibility by 9: A number is divisible by 9 if and only if its digital sum is divisible by 9.

Example: Is the number 873 divisible by 9?

- Add the digits: 8 + 7 + 3 = 18
- Since 178 is divisible by 9, 873 is also divisible by 9.

Divisibility by 3: A number is divisible by 3 if and only if its digital sum is divisible by 3.

Example: Is the number 567 divisible by 3?

- Add the digits: 5 + 6 + 7 = 18
- Since 18 is divisible by 3, 567 is also divisible by 3.

Divisibility by 11:

If the difference between the sum of digits at an even place and the sum of digits at the odd place is divisible by 11, then the whole number is divisible by 11

Ex; 54659, 968 etc.

Example 8: If the given number 925x85 is divisible by 11, then the smallest value of x is:

Solution:

Divisibility law of 11: If the difference of the alternating sum of digits of the number is a multiple of 11 (e.g. 2343 is divisible by 11 because 2 - 3 + 4 - 3 = 0, which is a multiple of 11).

The number 925x85 is divisible by 11,

Then 9 – 2 + 5 – x + 8 – 5 = 11

⇒ 15 – x = 11

⇒ x = 15 – 11

∴ The correct answer is x = 4.

Divisibility by 12:

The number which is divisible by both 3 and 4 is also divisible by 12.

Ex; 456, 588 etc.

Example 9: If a 4-digit number 273x is divisible by 12 and a 7-digit number y854z06 is divisible by 11, then what is the value of (x + y + z)?

Solution:

Given:

273x is divisible by 12

y854z06 is divisible by 11.

Concept:

A number is divisible by 12, if the given number is divisible by 3, and 4.

A number is divisible by 3, if the sum of the digits of the number is divisible by 3.

A number is divisible by 4, if the last two digit of the number is divisible by 4.

A number is divisible by 11, if the difference between the sum of the alternate digits of the given number is either 0 or divisible by 11.

Calculation:

In a four-digit number 273x , x should be 6.

As 2736 is the number which fulfills all the divisibility rules of 3 and 4.

7-digit number y854z06.

the sum of the alternate digits from right to left at odd place= y + 5 + z + 6

⇒ 11 + y + z

the sum of the alternate digits from right to left at even places = 8 + 4 + 0

⇒ 12

According to the divisibility rule of 11, the difference between the sum of the digits at odd places and sum of the digits at an even place is divisible by 11.

11 + y + z - 12 = 11

⇒ y + z - 1 = 11

⇒ y + z = 11 + 1

⇒ y + z = 12

value of x + y + z

⇒ 6 + 12

⇒ 18.

∴ The correct answer is (x + y + z) = 18.

Common divisibility rule for 7, 11, and 13:

To determine if a number is divisible by 7, 11, or 13, you can use the following method:

Separate the Number: Start from the rightmost digit of the number and separate it into groups of three digits each.

Calculate the Difference: Take the alternating sum of these three-digit groups, subtracting the sum of groups placed in odd positions from the sum of groups placed in even positions.

Check Divisibility: If the resultant difference is divisible by 7, 11, or 13, then the original number is also divisible by these numbers.

Example 10: Consider the number 100,014,915:

Solution:

Break it into groups of three from the right: 915, 014, and 100.

Calculate the alternating sum: 915 - 014 + 100 = 1001

Check divisibility: If 1001 is divisible by 7, 11, or 13, then so is 100,014,915.

This rule simplifies the process of checking divisibility for large numbers and is particularly useful when working with numbers that are difficult to factor directly.

Example 11: What is the largest 3-digit number divisible by 13?

Solution:

⇒ Largest 3 digits number = 999

⇒ Now we will divide 999 by 13

⇒ By dividing we will get 11 as a remainder

The largest 3 digit number divisible by 13 = 999 - 11 = 988

∴ The correct answer is 988.

Example 12: If a six digit number 738A6A is divisible by 11 and a three digit number 68X is divisible by 9, then the value of (A + X) is:

Solution:

Concept:

Divisibility of 11 = add digits placed on even place – add digits on odd place = 0 or 11

Divisibility of 9 = sum of all digits must be divisible by 9

Calculation:

Digits of even place = 7 + 8 + 6 = 21

Digits on odd place = 3 + A + A = 3 + 2A

Now subtract both the equations

⇒ 21 – 3 – 2A

⇒ 18 = 2A

⇒ A = 9

68X is divisible by 9

6 + 8 + X ⇒ 14 + X = 18

X = 4

(A + X) = (9 + 4)

(A + X) = 13

∴ The correct answer is 13.

Example 13: Find the largest number which is smaller than 10000 and it is exactly divisible by 7, 8, 12 and 14?

Solution:

Concept:

Find the LCM of the given numbers and then divide 10000 by LCM and finally, the remainder needs to be subtracted from 10000 to get the desired number.

Calculation:

LCM of 7, 8, 12 and 14 = 168

Divide 10000 by 168:

Quotient = 59

Remainder = 88

The nearest number which is smaller than 10000 and divisible by 168 = 10000 – 88 = 9912

∴ The correct answer is 9912.

Example 14: Find the smallest number that must be added to 79462, so that the sum is divisible by 6.

Solution:

Given:

The smallest number that must be added to 79462, so that the sum is divisible by 6.

Concept used:

When the number is divisible by both 2 and 3 then the particular number is divisible by 6.

Divisibility rule by 2: When the last digit of a number is either zero or even then the number is divisible by 2.

Divisibility rule by 3: When the sum of the digits of a number is divisible by 3 then the number is divisible by 3.

Calculations:

As per the divisibility rules of 3, adding the given number = 7 + 9 + 4 + 6 + 2 = 28

If we add 2 with 28 it will become 30 which is divisible by 3 as well as with 2 as the last digit is zero.

So, the smallest number which is to be added is 2.

∴ **The correct answer is 2.**

Example 15: The largest number of four digits that is exactly divisible by 17 and 36 is:

Solution:

Calculation :

LCM of 17 and 36

$\Rightarrow 17 = 17 \times 1$

$\Rightarrow 36 = 18 \times 2 = 2^2 \times 3^2$

LCM = $2^2 \times 3^2 \times 17 = 612$

The largest number of 4 digits will be 9999.

Now, 9999/612 = 16.38

If we multiply 612 by 16 we get the number as 9792.

∴ **The correct answer is 9792.**
